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Bantay-Baha

Trace the Pattern,
Target the Cure:
Advancing Public Health
Through Data Science

G is Cash

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EXECUTIVE SUMMARY

Eight to nine tropical cyclones cross the Philippines each year, and flood-linked disease follows within weeks: Typhoon Ondoy (2009) was followed by 2,089 leptospirosis cases and 162 deaths, and the 2019 dengue epidemic reached 146,062 cases by July. Response still begins after hospitals fill, and the communities most at risk are often the ones a storm has cut off from care.

Bantay-Baha ("flood watch") is a decision-support dashboard that tells the Department of Health (DOH) regional offices and local government unit (LGU) health officers, before or immediately after a typhoon, which provinces are most likely to see a disease surge and which of those will lose access to a hospital. It links three graphs: (1) a flood-exposure layer from satellite-observed Sentinel-1 flood extent, LiDAR Portal for Archiving and Distribution (LiPAD) hazard maps, and river-routed rainfall; (2) a causal lead-lag outbreak graph learned from province-week DOH dengue surveillance with Peter and Clark Momentary Conditional Independence (PCMCI+), which identifies sentinel provinces that rise first; and (3) a road-and-roll-on/roll-off (RoRo) ferry network that simulates storm-closed routes to measure lost hospital access. The first two feed a quantile gradient-boosted forecast of 1- to 4-week-ahead incidence, benchmarked against time-series foundation models and scored as international forecast hubs do. Surge probability and the population cutoff from care combine into one priority ranking with recommended action tiers.

We will backtest on Typhoons Ompong (2018), Tisoy (2019), Rolly (2020), and Ulysses (2020). No Philippine tool conditions outbreak risk on a specific storm and joins it with storm-specific loss of healthcare access; Bantay-Baha moves response from reacting to outbreaks to getting ahead of them.

KEYWORDS

Early warning system, dengue, leptospirosis, flooding, typhoon, graph analytics, causal discovery, probabilistic forecasting, healthcare access, Philippines

BACKGROUND

About 20 tropical cyclones enter the Philippine Area of Responsibility each year, and eight or nine cross the country, mostly between July and October (PAGASA, n.d.). Each storm brings flooding, which in turn brings disease. After Typhoon Ondoy in 2009, more than 2,089 people were treated for leptospirosis, with 162 dead, which is over five times the national leptospirosis deaths of 2008 (Balbuena et al., 2010). Manila hospital data show the leptospirosis risk from rainfall peaks two weeks later and is mediated by flooding (Matsushita et al., 2018). Dengue cases also rise with the rain. In 2019, the DOH declared a national epidemic after 146,062 cases and 622 deaths from January to July (The Philippine Star, 2019). Recent reviews call for a unified surveillance registry and tools that prioritize where to act (Interior et al., 2025; Labana, 2026).

The data needed to act early exist but are scattered across surveillance, hazard, climate, and facility datasets. Dengue forecasting from climate has been studied repeatedly in the Philippines, from province-level early warning (Buczak et al., 2014) to Metro Manila machine learning (Carvajal et al., 2018) and a nationwide comparison of eight forecasters (Olana et al., 2025). Flood-driven loss of hospital access has been modelled for Metro Manila, where a 15 cm flood leaves about 22% of residents without access to higher-level care (Tariverdi et al., 2023), and dengue source regions have been identified by synchrony analysis (van Panhuis et al., 2015). These dimensions have never been tied together, as no tool can tell how dengue risk might spike after a certain typhoon, show how storms cut off access nationwide (including ferry routes), or combine the two into one ranking for health officers. The problems share a structure that graph methods model directly, which is that water flows along rivers, outbreaks move between connected places, and patients travel along roads and ferries.

THE PROBLEM

Philippine health authorities currently have no tool that anticipates, before or immediately after a typhoon, where flood-linked disease surges will occur and which affected communities will be cut off from care. Prophylaxis, vector control, and medical teams are deployed after cases rise, and isolated communities are reached last. Existing dengue models are not storm-conditioned, provide point forecasts without calibrated uncertainty, and disregard whether patients are able to reach a hospital. Likewise, hazard tools such as UP Nationwide Operational Assessment of Hazards (NOAH) carry no health layer.

OBJECTIVES

By the PJDSC 2026 final-round submission, the team will:

- Assemble a province-by-week panel for all 81 provinces from 2016 to 2020, joining DOH Epidemiology Bureau (DOH-Epi) dengue cases, Integrated Multi-satellite Retrievals (IMERG) rainfall, Sentinel-1 flood fraction, LiPAD hazard share, and International Best Track Archive for Climate Stewardship (IBTrACS) typhoon exposure on PSGC codes.
- Learn a lead-lag outbreak graph with PCMCI+, identify sentinel provinces (PageRank) and communities (Louvain), and test whether sentinel features reduce follower-province forecast error out of sample.
- Forecast 1- to 4-week-ahead dengue incidence per province with a quantile gradient-boosted ensemble, achieving a relative Weighted Interval Score (WIS) below 1.0 against the seasonal-naive baseline and 90% interval coverage of 80 to 95% on rolling-origin evaluation over 2018 to 2019.
- Quantify storm-specific loss of hospital access for every city and municipality with a friction-surface and a road-plus-RoRo network method for Ompong, Tisoy, Rolly, and Ulysses, with sensitivity bounds and the Metro Manila benchmark of Tariverdi et al. (2023).
- Deploy a dashboard that returns a ranked province list with calibrated risk, access loss, sentinel status, drivers, and an action tier for any historical or shifted

SCOPE & LIMITATIONS

This study is nationwide in scope and covers all 81 provinces at weekly resolution for 2016 to 2020, which matches the DOH-Epi dengue file on Humanitarian Data Exchange (HDX). The access analysis is conducted per city and municipality and is summarized per province as the share of the population that is cut off. Dengue is the primary disease of interest because consistent subnational data exist for it. Leptospirosis is considered a stretch goal since no open subnational series exists, so it enters as a national weekly lag analysis and, if DOH weekly surveillance reports (2022 to 2026) are digitized in time, as a region-week series. Storm scenarios will use historical IBTrACS tracks and shifted variants. The intended users are DOH regional offices, LGU health offices, and local Disaster Risk Reduction and Management (DRRM) offices.

As with all studies, the proposed work also has some limitations. Surveillance data carry reporting delays and under-reporting, and COVID-19 collapsed 2020 reporting, so 2020 is used for access scenarios only. Sentinel-1 may fail to detect urban and vegetated floods and can miss the flood peak. OpenStreetMap (OSM) completeness outside Luzon is uneven, and for this reason two access methods are reported. It should also be noted that PCMCI+ links are statistical dependencies and not proof of transmission. Finally, the tool is meant to support decisions and does not replace clinical or field judgment; it uses only aggregated, non-identifiable data.

METHODOLOGY

Bantay-Baha builds three graphs, turns each of them into features for a single province-week table, and combines a calibrated disease forecast with storm-driven access loss into one priority ranking (Figure 1, Appendix A; data sources in Table 1, Appendix B).

Layer 1: Flood exposure (prevent)

For each typhoon, Sentinel-1 radar before-and-after change detection in Google Earth Engine (UN Platform for Space-based Information for Disaster Management recommended practice) yields the flooded fraction and flooded population (WorldPop 100 m) per province and municipality, cross-checked against Copernicus Emergency Management Service delineations (EMSR312 Ompong, EMSR478 Rolly). Static susceptibility is the share of the population inside LiPAD 5- and 25-year flood hazard polygons. Dynamic forcing uses GPM IMERG 3- and 7-day rainfall and rainfall accumulated downstream along the HydroRIVERS network, so towns near river mouths register upstream rain.

Layer 2: Causal lead-lag outbreak graph (stop outbreaks early)

Province-week incidence is log-transformed and seasonally differenced so shared rainy-season peaks do not create false links. PCMCI+ (Runge, 2020) in Tigramite, with partial-correlation tests and lags up to three weeks, recovers a directed lead-lag graph with bootstrap confidence per edge; lagged cross-correlation is the baseline graph. PageRank ranks sentinel provinces that rise first; Louvain groups provinces that move together for coordinated response. Graphs are learned only inside each training window to avoid leakage.

Forecast model

A pooled Extreme Gradient Boosting (XGBoost) model with province as a categorical feature predicts one- to four-week-ahead incidence from lagged cases, calendar week, lagged rainfall, flood exposure (Layer 1), and lagged cases of PCMCI+ leader provinces (Layer 2), trained with the multi-quantile

objective at the 23 levels used by the CDC FluSight hub and explained with SHapley Additive Explanations (SHAP). An identical model without graph features is the ablation. Zero-shot baselines set the bar: Chronos-2 (Ansari et al., 2025) with rainfall and flood covariates, Tabular Prior-data Fitted Networks—Time Series (TabPFN-TS), a seasonal-naive forecast, and a fixed-adjacency Message Passing Neural Network + Long Short-Term Memory (MPNN-LSTM). The final forecast is the median ensemble, with conformal (EnbPI) intervals as a check. Models are trained on earlier years and are evaluated by rolling-origin reforecasts, never a random split, and scored with the Weighted Interval Score (Bracher et al., 2021), relative WIS, and coverage.

Layer 3: Access graph and storm scenarios (strengthen healthcare reach)

Nodes are city and municipality centers and hospitals from the DOH National Health Facility Registry and OpenStreetMap. Edges are OSM road links and the 108 province-to-province RoRo routes in the MARINA inventory (RORPH), weighted by travel time and frequency. A multi-source shortest-path search gives each town's travel time to the nearest hospital. For each typhoon, ferry routes inside the IBTrACS gale-radius buffer are removed, and road links inside the observed flood polygon are slowed or cut; towns with no remaining path are flagged. A raster method, the Malaria Atlas Project friction surface (Weiss et al., 2020) with flooded cells impassable, runs in parallel following Petricola et al. (2022); the disagreement between methods is reported as uncertainty.

Priority score and product

For a chosen typhoon, each province's priority is P = R x A, where R is the forecast probability that incidence over the next four weeks exceeds the province's 90th percentile, and A is the share of the population cut off from hospital care, so P reads as the expected share of residents at elevated risk who cannot reach care. The Streamlit dashboard takes a historical or shifted track and returns a priority map, a risk-versus-access 2x2, a ranked table with predicted cases and intervals, cut-off population, sentinel status and SHAP drivers, and an action tier: monitor sentinel sites, pre-position doxycycline and vector-control supplies, or deploy mobile clinics before roads and ferries close. Scenarios are pre-computed, so results return in under one minute.

Validation

We backtest the forecast on Ompong (2018), Tisoy (2019), and the 2019 season, and the access module on all four typhoons, checking whether high-priority provinces saw higher incidence in the following four weeks. Sensitivity to travel speed, ferry rules, and storm buffer is reported as rank stability (Kendall's tau). Three target users complete a 15-minute scenario task and the System Usability Scale.

Tools

Python with pandas, GeoPandas, Network eXplore (NetworkX), OpenStreetMap + NetworkX (OSMnx), Tigramite, XGBoost, chronos-forecasting, tabpfntime-series, Model Agnostic Prediction Interval Estimator (MAPIE) and SHAP, Google Earth Engine, Streamlit, and all code in a public GitHub repository with a one-command rebuild.

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APPENDICES

Appendix A. Figure 1. Bantay-Baha pipeline: three graphs, two models, one ranking

